

In TypeScript, a **module** is simply a **file that contains code that can be shared with other files** using `export` and `import`.

Think of a module as a **separate container of related code**.

Simple example

Suppose you have:

```
project/  
|  
├─ math.ts  
└─ app.ts
```

math.ts is a module:

TypeScript

```
export function add(a: number, b: number): number {  
    return a + b;  
}  
  
export function subtract(a: number, b: number): number {  
    return a - b;  
}
```

app.ts imports the module's functionality:

TypeScript

```
import { add, subtract } from "./math";  
  
console.log(add(10, 5));  
console.log(subtract(10, 5));
```

So:

```
math.ts  
|  
| export  
↓  
app.ts  
|  
| import  
↓  
uses add() and subtract()
```

Why do we use modules?

Imagine putting your entire Playwright framework into one huge `.ts` file:

```
Login code
Home page code
Insurance code
Database code
Utilities
Test data
Test cases
...
```

It would become difficult to maintain.

Instead, we separate the code:

```
project/
|
├─ pages/
|   ├── LoginPage.ts
|   └── HomePage.ts
|
├─ utils/
|   ├── DateUtils.ts
|   └── TestDataUtils.ts
|
└─ tests/
    ├── login.spec.ts
    └── insurance.spec.ts
```

Each `.ts` file can act as a **module**.

For example:

LoginPage.ts

TypeScript

```
export default class LoginPage {

  login(username: string, password: string) {
    console.log("Login successful");
  }
}
```

login.spec.ts

TypeScript

```
import LoginPage from "../pages/LoginPage";

const loginPage = new LoginPage();

loginPage.login("admin", "password");
```

Here:

- `LoginPage.ts` → **module**
- `export default` → makes `LoginPage` available outside the module
- `import` → brings `LoginPage` into another module
- `login.spec.ts` → another **module**

Module vs class

These are different concepts:

Concept	Meaning
Module	A file used to organize/share code
Class	A blueprint for creating objects
Function	A reusable block of code
Export	Makes something available outside the module
Import	Brings something from another module

Easy definition for your notes

A TypeScript module is a file containing related code that can expose selected variables, functions, classes, or types to other files using `export`, and consume them using `import`.

This concept becomes especially important in **Playwright**, because Page Objects, utilities, test data, and configuration are typically organized into separate modules.